

Day – 12 Other Protocols in Action Theory.

Task – 1 Install Filezilla server.

- However, setting up FileZilla Server on a Windows machine is a straightforward and classic way to test FTP (File Transfer Protocol) file sharing. Here is the step-by-step guide to setting it up, creating a user, and sharing a folder:
-
- **Step 1: Download and Install FileZilla Server**
- Download the official installer for FileZilla Server (Windows) from the source website.
- Run the installer and go through the setup wizard:
- Leave the components at their default settings.
- Choose whether to install the server as a service that starts automatically with Windows (recommended).
- Set an administrative password when prompted to secure the server management interface.
-
- **Step 2: Open the Administrative Interface**
- Launch the FileZilla Server Interface (or use the web-based admin interface introduced in modern versions).
- Connect to the server instance using your admin credentials (default port is usually 14147 for local administration, or localhost).
-
- **Step 3: Create a New User Account**
- In the FileZilla Server interface, navigate to Edit > Users (or open the Users configuration menu).
- Click Add to create a new user account (e.g., ftpuser).
- Check the box to Require a password and enter a secure password for this user account.
-
- **Step 4: Set Up and Map a Shared Folder**
- While still in the User settings menu, look for the Shared folders section on the left-hand panel.
- Click Add and choose a folder on your computer that you want to share (for example, create a folder named C:\FTPShare).
- Set the appropriate Permissions for the user:
- Files: Read, Write, Delete, Append.
- Directories: Create, Delete, List, Subdir list.

- Designate this folder as the Home directory (marked with a H) so the user lands straight in this directory upon logging in.
- Click OK to save the user and folder settings.
-
- Step 5: Test Your FTP Connection
- You can now test your new FTP server locally using a command-line client or FileZilla Client:
- Open your Windows Command Prompt (cmd).
- Type ftp 127.0.0.1 and press Enter.
- Log in with the username (ftpuser) and the password you created.
- Type ls or dir to list the files in your newly shared folder!

Task – 2 Connecting client via FTP server.

- Here is how you can perform this test on your local machine using FileZilla Client:
-
- Step 1: Open FileZilla Client and Enter Connection Details
- Open FileZilla Client on your computer.
- In the Quickconnect bar at the very top of the window, fill in your server credentials:
- Host: 127.0.0.1 (or your local machine's IP address if testing across two devices)
- Username: ftpuser (the account you created earlier)
- Password: [The password you assigned to that user]
- Port: 21 (the standard FTP control port)
- Click the Quickconnect button (or press Enter).
-
- Step 2: Observe the Connection Log
- Watch the message log right above the file windows. You will see a sequence like this:
- Connecting to 127.0.0.1:21...
- Connection established, waiting for welcome message...
- 220 Welcome to FileZilla Server
- USER ftpuser -> 331 Password required
- PASS ***** -> 230 Logged on
- Status: Directory listing of / successful
-
- Step 3: Locate and Upload the Image File
- Once connected, the FileZilla interface splits into two main sections:
- Left Side (Local Site): This displays your computer's local hard drives and folders. Browse through your directories until you find any image file (for example, a .jpg or .png file saved on your Desktop).
- Right Side (Remote Site): This displays your server's shared folder (C:\FTPShare).

- To upload the image:
- Simply drag and drop the image file from the left panel (Local site) directly into the right panel (Remote site).
- Alternatively, you can double-click the image file on the left, and FileZilla will automatically queue and transfer it to the server.
-
- Step 4: Verify the Successful Transfer
- Look at the bottom pane of FileZilla under the Successful transfers tab. You should see your image file listed with a transfer status of 1.
- Go to C:\FTPShare on your server machine—you will see the image file sitting right there, successfully uploaded via FTP!

Task – 3 FTP server transferring.

- Here is how you can perform the download and verify that the file integrity remains intact:
-
- Step 1: Navigate to a Destination Folder on Your Local Machine
- In the Left Panel (Local site) of FileZilla Client, browse away from your original folder.
- Select a completely different folder (for example, create a new folder on your Desktop named FTP-Downloads and open it) so you can clearly see the incoming file.
-
- Step 2: Locate and Download the File from the Server
- In the Right Panel (Remote site), locate the image file you uploaded earlier sitting inside your server's shared folder (C:\FTPShare).
- Drag and drop the image file from the right panel directly into the left panel (into your new FTP-Downloads folder).
- *Alternative:* You can double-click the file on the right, and FileZilla will automatically queue and transfer it to your currently active local directory.
-
- Step 3: Verify the Successful Transfer and File Size
- Check the bottom pane of FileZilla under the Successful transfers tab to ensure the download completed with a status of 1 and zero errors.
- Go to your local file explorer on your computer:
- Right-click the original image file and select Properties (note the exact file size in bytes).
- Right-click the newly downloaded image file in your FTP-Downloads folder and select Properties.

- Verification: Confirm that the file sizes match down to the exact byte. This ensures that the FTP download transferred the binary data accurately without corruption.

Task – 4 List the three scenario SNMP is used.

- Here are three real-world scenarios where SNMP plays a vital role in network management:
-
- 1. Internet Service Provider (ISP) Bandwidth and Traffic Monitoring
- The Scenario: An ISP managing thousands of residential and commercial broadband connections needs to ensure its core routing and switching infrastructure isn't getting choked by traffic bottlenecks.
- How SNMP Works Here: Network management software (like PRTG or LibreNMS) continuously polls core routers using SNMP. It queries specific object identifiers (OIDs) to track interface metrics—such as bits-in/bits-out, packet error rates, and CPU utilization. If a particular trunk line reaches 85% capacity, SNMP triggers an automated warning so engineers can upgrade the link before users experience latency.
-
- 2. University Campus Wi-Fi Access Point (AP) Health & Load Tracking
- The Scenario: A sprawling university campus has hundreds of wireless access points distributed across lecture halls, dorms, and libraries, serving thousands of simultaneous student devices.
- How SNMP Works Here: The campus network operations center (NOC) uses SNMP to monitor the operational status of every access point. It tracks metrics like memory usage, temperature sensors, and the total count of associated client devices per AP. If an access point's CPU spikes or it suddenly drops offline, SNMP sends an alert so the IT team can reboot or replace it.
-
- 3. Enterprise Data Center Hardware and Environmental Alerts (SNMP Traps)
- The Scenario: A large enterprise data center running critical web services cannot afford unexpected hardware failures or server room overheating.
- How SNMP Works Here: Instead of just polling, data center equipment (servers, rack switches, UPS battery backups, and AC units) uses SNMP Traps. When an event occurs—such as a power supply failure, a spinning fan dropping below safe RPMs, or ambient server room temperature crossing a threshold—the device *proactively pushes* an alert (a trap) to the central monitoring station. This enables immediate physical intervention before a catastrophic outage occurs.

Task – 5 Difference between FTP and HTTP.

- 1. Port Number
- FTP: Uses two distinct ports—Port 21 is used for sending control commands and establishing the connection, while Port 20 (or a dynamic range in passive mode) is used for actual data transfer.
- HTTP: Uses Port 80 by default for standard web traffic (or Port 443 for secure HTTPS connections).
-
- 2. Security
- FTP: Traditional FTP transmits both authentication credentials (usernames and passwords) and file data in plain text, making it vulnerable to packet sniffing unless upgraded to secure alternatives like SFTP or FTPS.
- HTTP: Standard HTTP also transmits data in plain text, but modern implementations rely heavily on HTTPS (with TLS encryption) to secure traffic between the client and server.
-
- 3. Data Transfer Method
- FTP: Relies on a dual-channel architecture, separating the control connection from the data connection to manage large file uploads and downloads efficiently.
- HTTP: Uses a request-response model over a single TCP connection, where the client sends a request (like a GET or POST) and the server responds with the requested resource or data payload.
-
- 4. Use Cases
- FTP: Designed specifically for bulk file management—such as uploading website files to a hosting server, backing up large databases, or sharing documents between users.
- HTTP: Designed for web browsing and application delivery, serving HTML pages, images, scripts, and APIs to web browsers and mobile applications.
-
- 5. Authentication
- FTP: Requires an explicit login session with a username and password (or anonymous access) before any file operations can take place.
- HTTP: Inherently stateless; it does not maintain a continuous login session at the protocol level. Authentication is typically handled at the application layer using session cookies, tokens, or header-based credentials.
-